

Unit - 5 Chemical method of analysis

Q.1 Explain with diagram the construction & working of high power liquid chromatography (HPLC)

→ High power liquid chromatography (HPLC) is a separation technique in which a liquid or properly dissolve solid sample is passed through a column at high pressure.

Principle:

The basic principle of HPLC is that it separates a sample into its constituents parts based on the relative affinities of distinct molecules for the mobile phase and the stationary phase used in the separation.

It is an analytical technique. use to separate, identify or quantify each component in a mixture the mixture is separated using basic principle of column chromatography and then identify & quantified by spectroscopy.

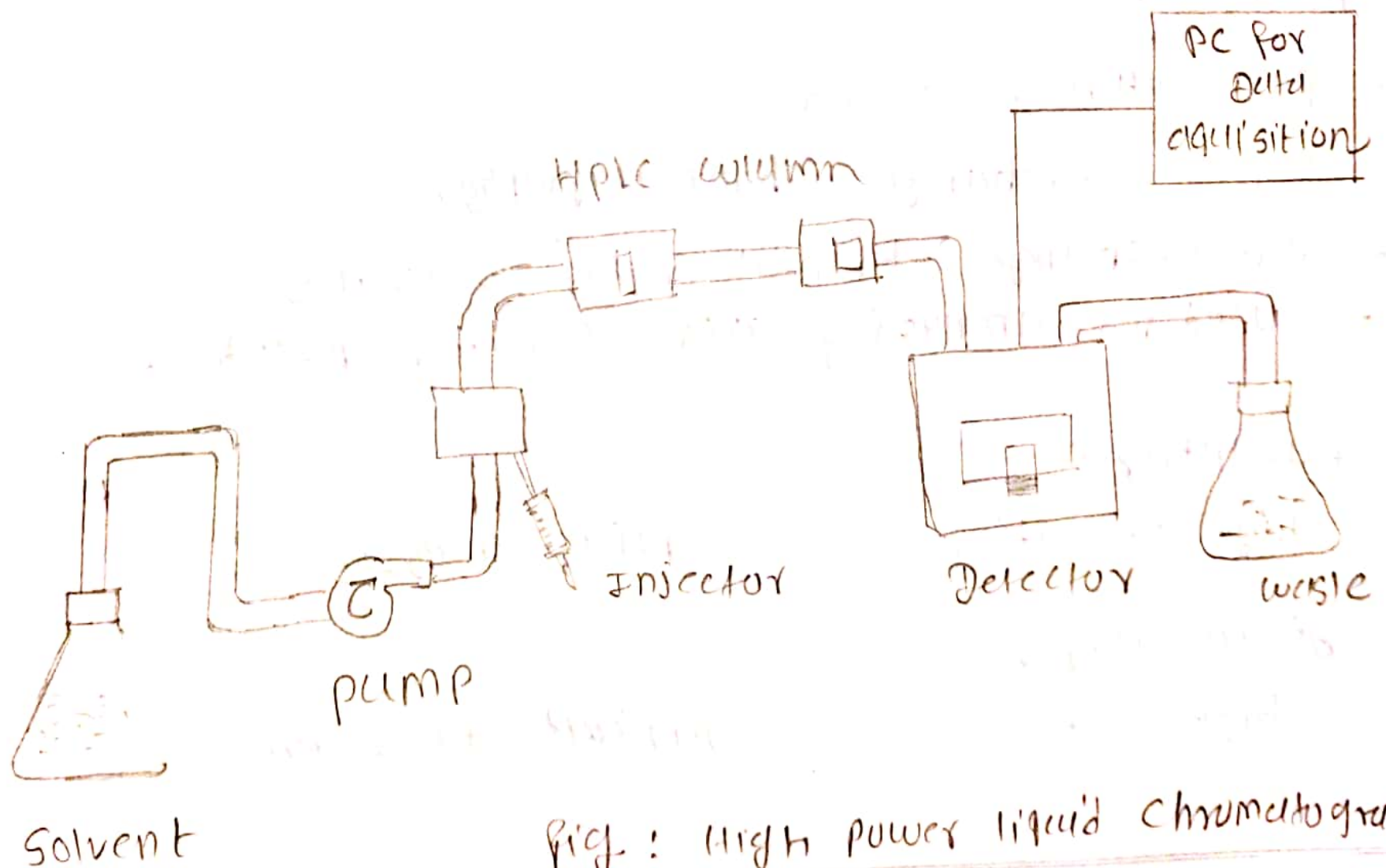


Fig: High power liquid chromatography.

- The purification takes place in a separation column between a stationary and mobile phase
- The stationary phase is granular material with very small porous particle in a separation column
- The mobile phase, on the other hand, is a solvent or solvent mixture which is forced at high pressure through the separation column.
- After leaving the column the individual substances are detected by a suitable detector and pass on as a signal to the HPLC software on the computer.
- At the end of this operation / run, a chromatogram in the HPLC software on the computer is obtained
- The chromatogram allows the identification and quantification of the different substances.

Applications:

- purification of water.
- Ligand - exchange chromatography.
- Ion-exchange chromatography of proteins.
- used for analyzing air & water pollutants.

Advantages

- high resolution
- quick analysis

Disadvantages

- high cost
- difficult to operate

Explain principle with diagram construction & working of pH meter.

A pH principle:- A pH meter is used to determine acidity or alkalinity of the solution. pH is the conc. of H^+ ions in the solution. A solution containing more H^+ ions remains acidic while the soln containing more OH^- ions remains alkaline.

It is necessary to determine concentration of acid and bases in the application like food industry, medical industry & other chemical industries. This can be done by measuring pH of sample solution.

pH of an aqueous solution is define as negative logarithm of its H^+ ions concentration i.e. $pH = -\log[H^+]$

pH scale:

Construction & working

Electronic pH measurement system consist of a measuring electrode, a reference electrode & potential measuring system.

① measuring electrode. :-

This electrode is dipped in the measuring solution so that potential is develop at the platinum electrode which is proportional to the pH of the measuring solution. This potential is measure by completing the circuit with the reference electrode.

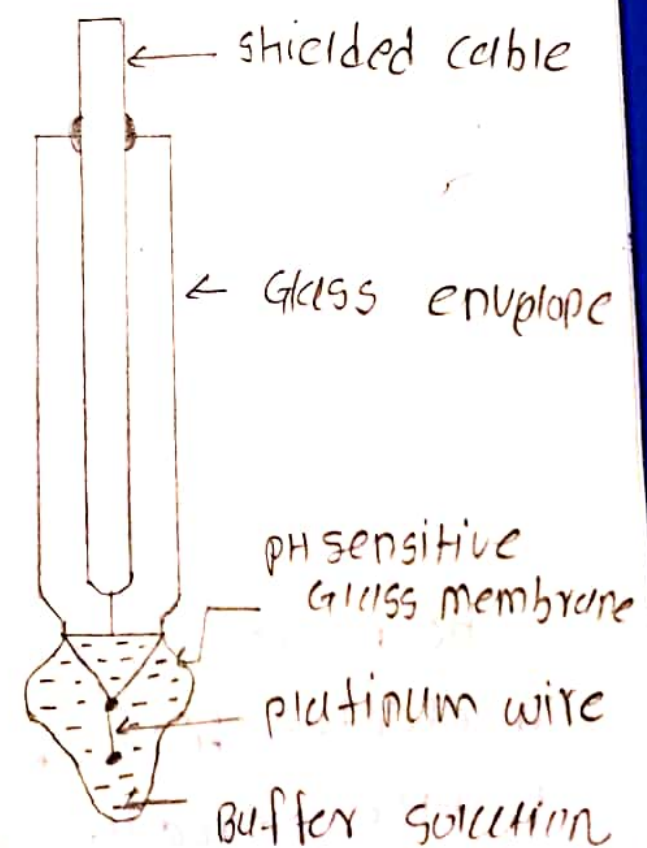


Fig. pH measuring glass electrode

Reference electrode:-

calomel electrode has glass envelope that contains glass tube which contains calomel solution along with platinum wire dipped in it. Thus tube is surrounded by KCl soln that slowly diffused or leaks into process liquid through liquid junction provided by asbestos fibre tip. Due to this reference electrode develops constant potential.

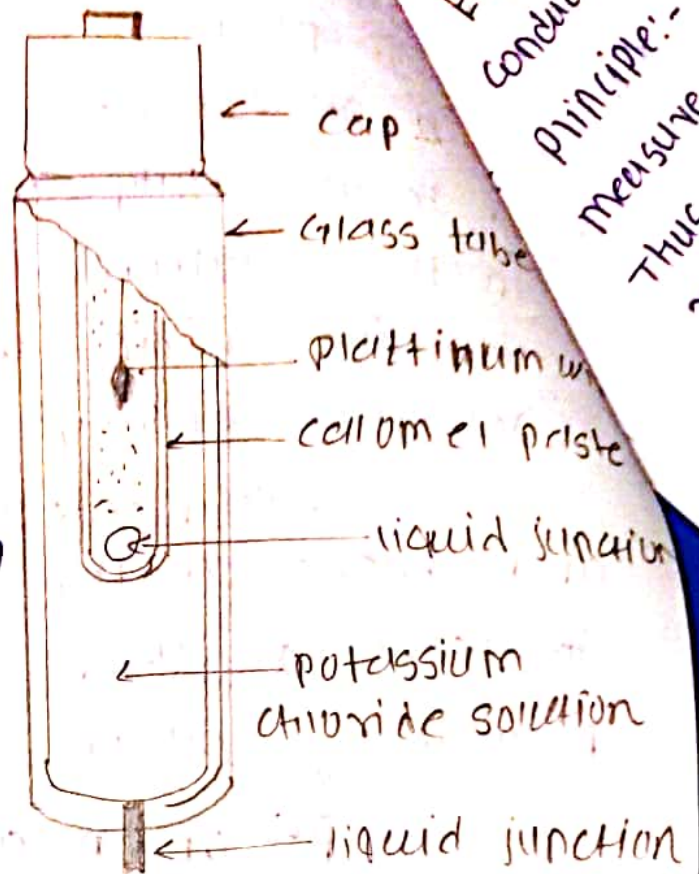


Fig: calomel electrode

Potential measuring system:-

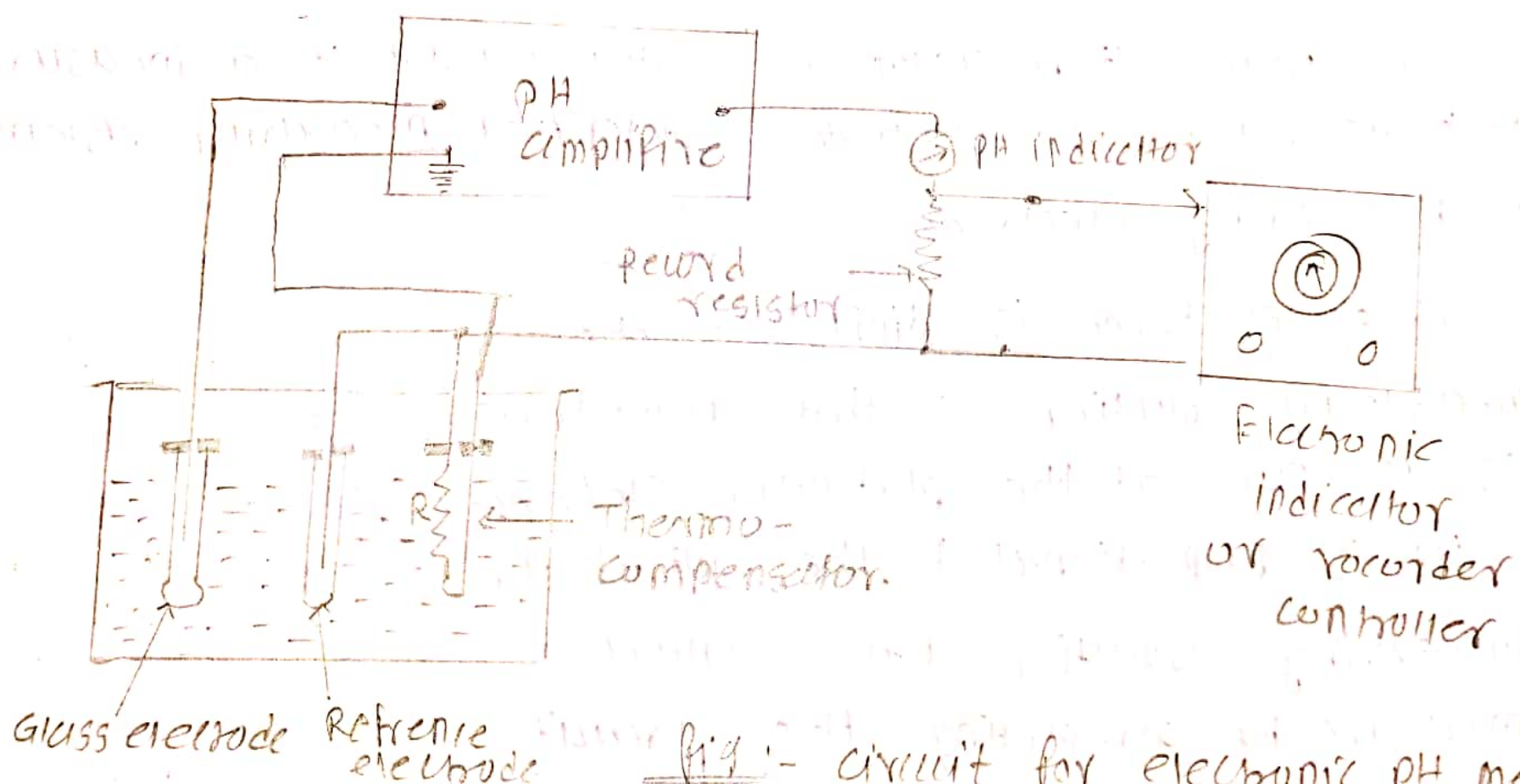


Fig: circuit for electronic pH measurement.

The measuring of the reference electrode together form an electrolytic cell whose output equals the sum of the voltage produced by the two electrodes. This net voltage is applied to a null-balance millivolt potentiometer in which the slide wire can be calibrated in terms of pH of measuring liquid.

Explain with diagram construction & working of conductometric meter.

principle:- The conductivity of an electrolyte is the measure of the ability of the solution carry electric current. Thus the conductance of the solution of electrolyte is related to the concentration of electrolyte.

The reciprocal of resistance R of electrolytic solution ($1/R$) is called the conductance.

$$\frac{1}{R} = k \left(\frac{a}{l} \right)$$

where k is the specific conductance in $\text{ohm}^{-1} \text{cm}^{-1}$

construction & working.

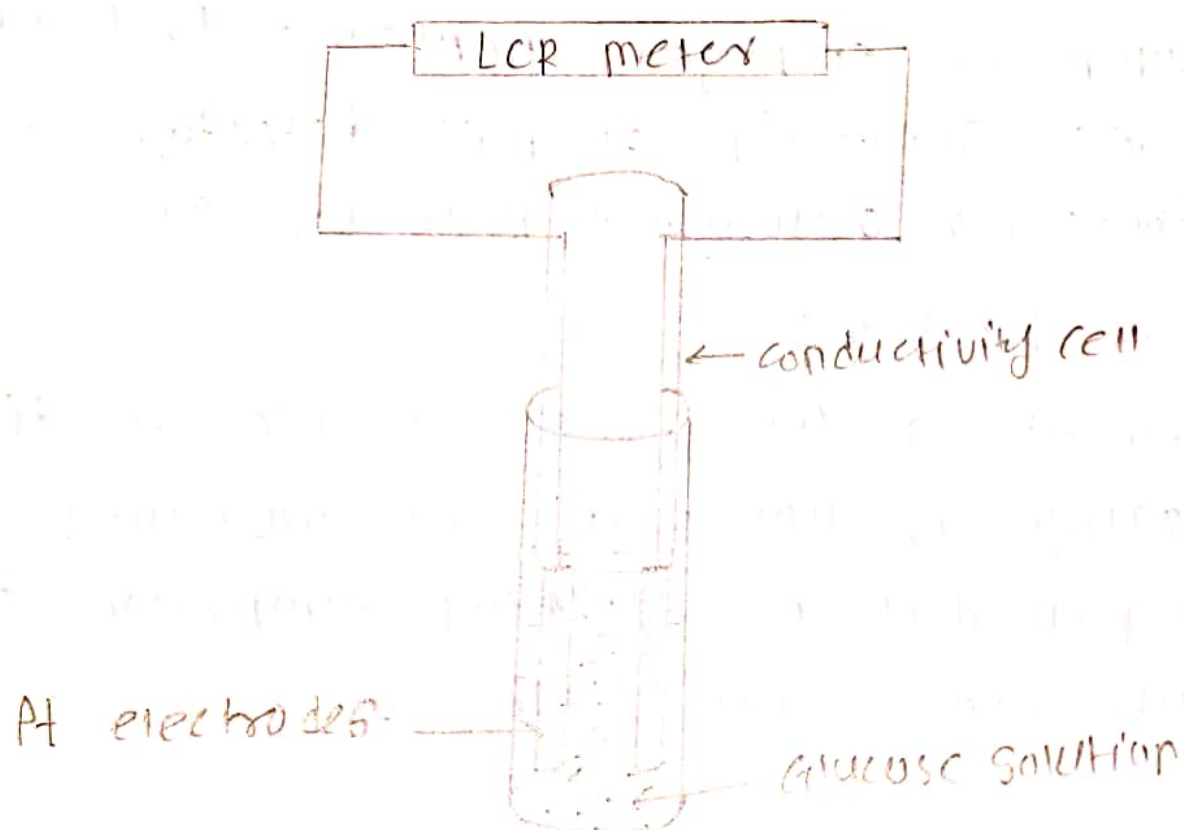


fig - conductometer.

The container with sample is placed betⁿ the plates of a capacitor, forming a part of the high frequency generator circuit. any change in composition of the solution results in changes in plate and grid current and voltage due to change in conductance & capacitance of the cell. The frequency of the parallel resonant circuit is given by

$$f = \frac{1}{2\pi} \sqrt{LC}$$

The advantages of placing the electrodes outside the container and out of direct contact with it is offered by this method. This eliminates the possibility of electrolysis or electrode polarisation.

Methods of measurement of conductance:

① Null method:

Conductivity is determined by measuring the resistance of a column of solution using Wheatstone bridge, where the conductivity cell forms one of the arms of the bridge. As the conductivity cell contains electrodes separated by a dielectric, an appreciable cell capacitance is invariably present. The commonly employed frequency is 1000 Hz.

Due to less sensitivity of null detector, use of magic eye or cathode ray oscilloscope is more popular.

② Direct reading method:-

In this the necessity of converting resistance readings into conductance readings is eliminated. The unbalanced bridge current is amplified in an electronic amplifier and displayed on a calibrated panel meter.

construction & working of gas chromatography with diagram.

Principle: Gas chromatography is based on the principle of partition (differential distribution) of an volatile compound (gas) in two phase-

a liquid phase covering the adsorbent surface and a gaseous phase of the eluting gas. with a fixed set of parameters.

(i.e. length & temp of the column, Nature & flow rate of the eluting gas).

Construction & working

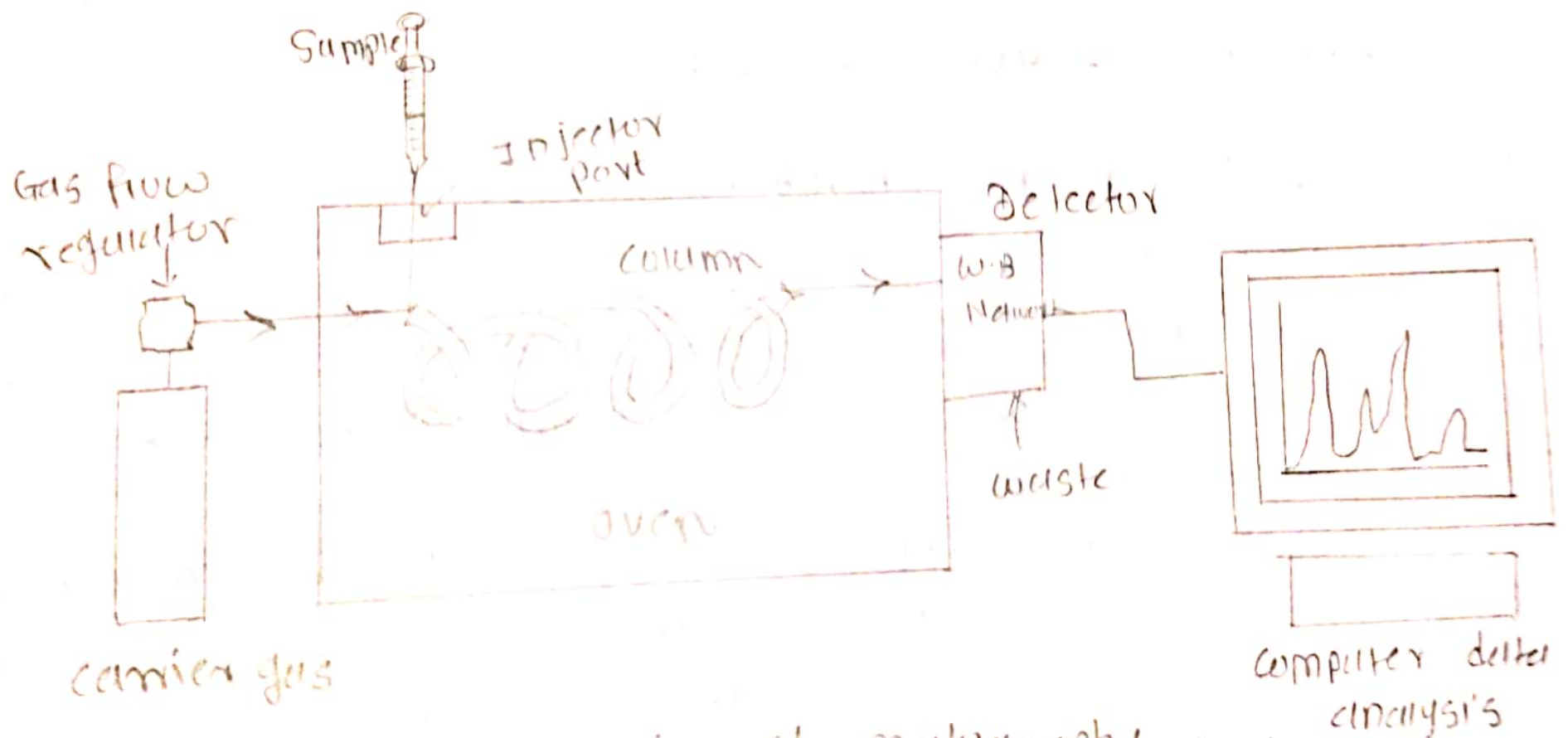


Fig. Gas chromatography analyzer

~~construction~~ above fig shows the set up for gas chromatography analyzer which consist of carrier gas bottle, chromatography column, reference & measuring thermal conductivity cells, connected in bridge circuit and fraction collection, measurement facility.

carrier gas: it just forces the sample gas through the column. Hence it may be any gas that does not react with sample. Usually He, H_2, N_2 or air are used as carrier gas.

Sample injection: The sample gas to be analysed is injected at a point as shown in figure. that flows the column along with carrier gas. A flash ~~evaporator~~ heater is provided at this point to vaporise the sample.

Separation column: - For Gas-liquid chromatography, usually partition type columns are used which consists of an inert solid material of small mesh coated with high boiling point liquid. This liquid coating absorbs the sample gas constituents at different rates.

measuring & reference chamber: -

These chambers have thermistor as sensing device. The carrier gas alone passes through reference chamber whose thermal conductivity decides the temperature and hence the resistance of thermistor bed.

Hence the difference in resistance of measuring and reference bed is only due to the separated components of the mixture. Hence the detector readings can be calibrated in terms of concentration of the separated component.

Indicator or recorder: -

The output of the bridge circuit is amplified and it is fed to the strip chart recording instrument.

By comparing the height of each peak with that produced by passing a calibrated sample of known composition over the measuring filament the percentage by volume of each constituent of sample gas can be determined.

- It is used for

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Principle: The basic principle of mass spectroscopy is to generate ions from either organic or inorganic compounds by any suitable methods, to separate these ions by their mass-to-charge ratio (m/z) and to detect them qualitatively and quantitatively.

construction and working

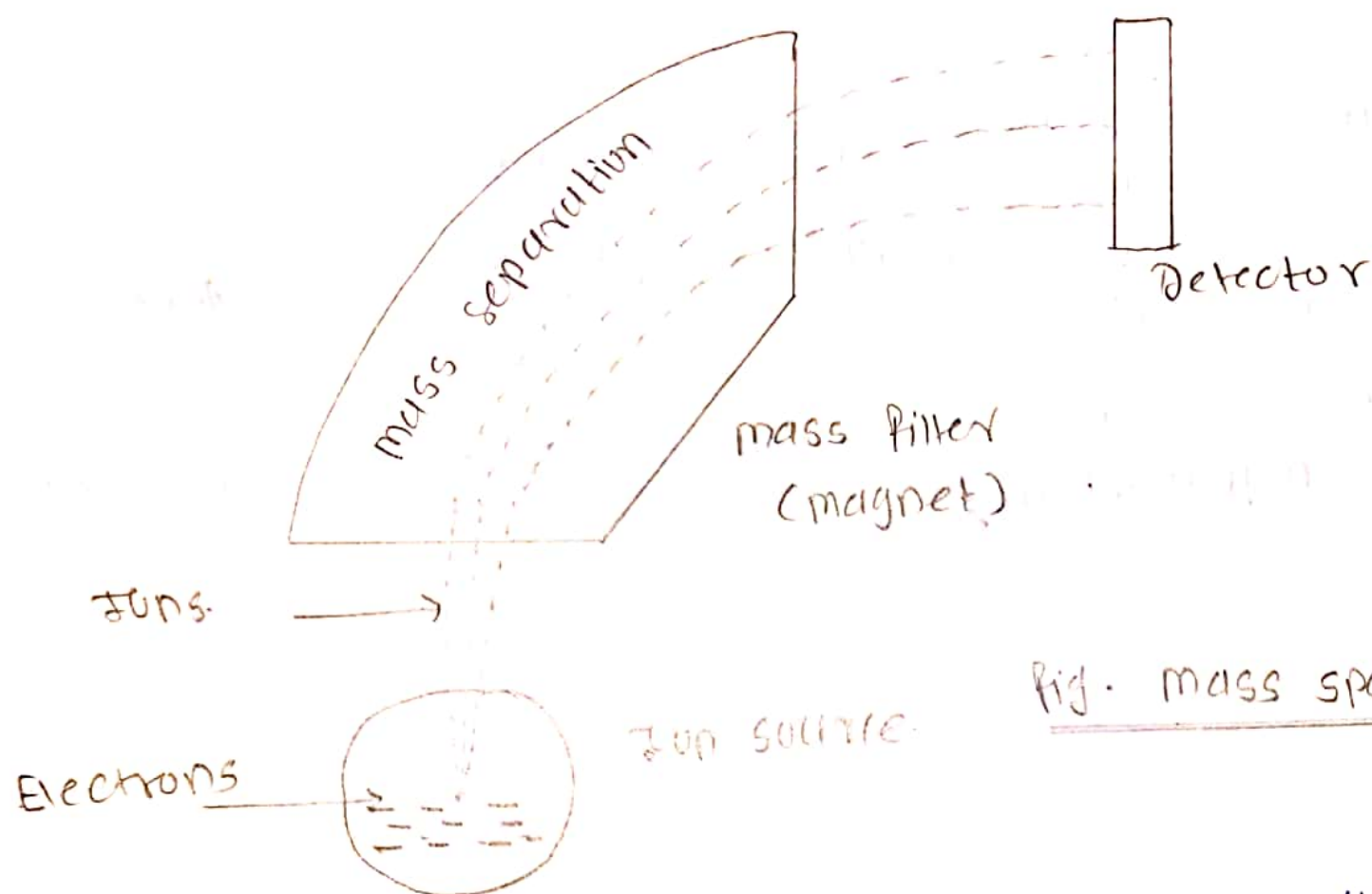


Fig. mass spectrometer.

- In this method a sample substance, usually a vapour of gas is bombarded by an electron beam in an evacuated chamber.
- This causes "knocking off" of the electrons from each of the different types of atom present in the substance. Such atoms from which electrons are knocked off are called as ions.
- These ions are then accelerated into a circular path by a magnetic field.

- The radius of the circular path varies with the mass of the ions thus ions are sorted into beams.
- The spectrum of ions is compared to the spectrum of the pure substance and mole fraction of the components are obtained by direct proportion.
- Thus spectrometer can separate masses differing by one part in 300 to one part in 1200. Hence mass spectrometer can be used for analysing the isotopes.

Application:

- Aliphatic and cyclic hydrocarbon compounds.
- light gases, olefins, paraffins, wet gases, propylene in petroleum refining.
- organic and inorganic gases and vapours in chemical manufacturing.

the beams. the spectrum of the

principle: when the frequency of specific vibration is equal to the frequency of IR radiation directed on the molecule, the molecule absorbs the radiation.

The IR portion of the electromagnetic spectrum extends from $7500 \text{ \AA}$ to $10^7 \text{ \AA}$. The manner in which electromagnetic radiation interacts with matter is a function of the wavelength of the radiation. IR radiation in the wavelength range $30,000$ to $10^5 \text{ \AA}$ have sufficient energy to excite molecular vibration and hence they can use for the detecting presence of molecular species.

Construction & working

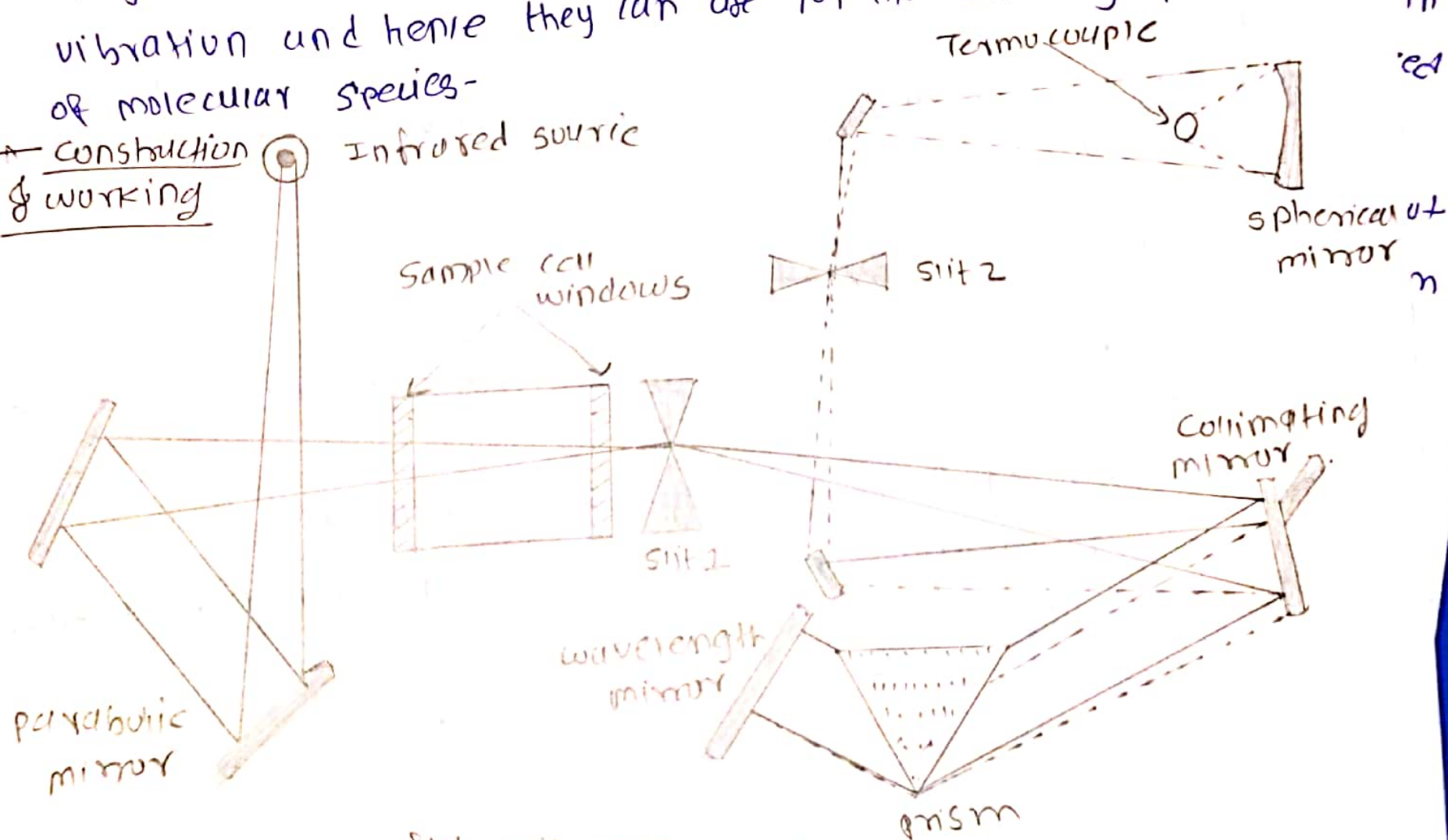


Fig - IR Spectrometer

— above fig shows the optical system of IR spectrometer. The IR radiation is produced by an electrical heated resistor. The IR beam passes the plane mirror then to parabolic mirror which is then falls on the slit through the sample to be analysed. The beam then passes to collimating mirror where it is made parallel.

- This parallel beam passes through the prism it is refracted.
- The wavelength mirror reflects the beam back through the prism where it is once again refracted.
- The wavelength of IR radiation receive by the sample is determine by the angle of setting of the wavelength mirror.
- Thus by rotating the prism the intensity of the each wavelength can be measure.
- The concentration of various components of the sample substance can be determine from the amount of absorption at any wavelength which is based on the Beer's law.

Applications

- To monitor CO_2 in the manufacturing of ethylene oxide and ammonia.
- monitoring of explosive and toxic hazards.
- Acetylene monitoring in the manufacturing of acetylene & vinyl chloride.

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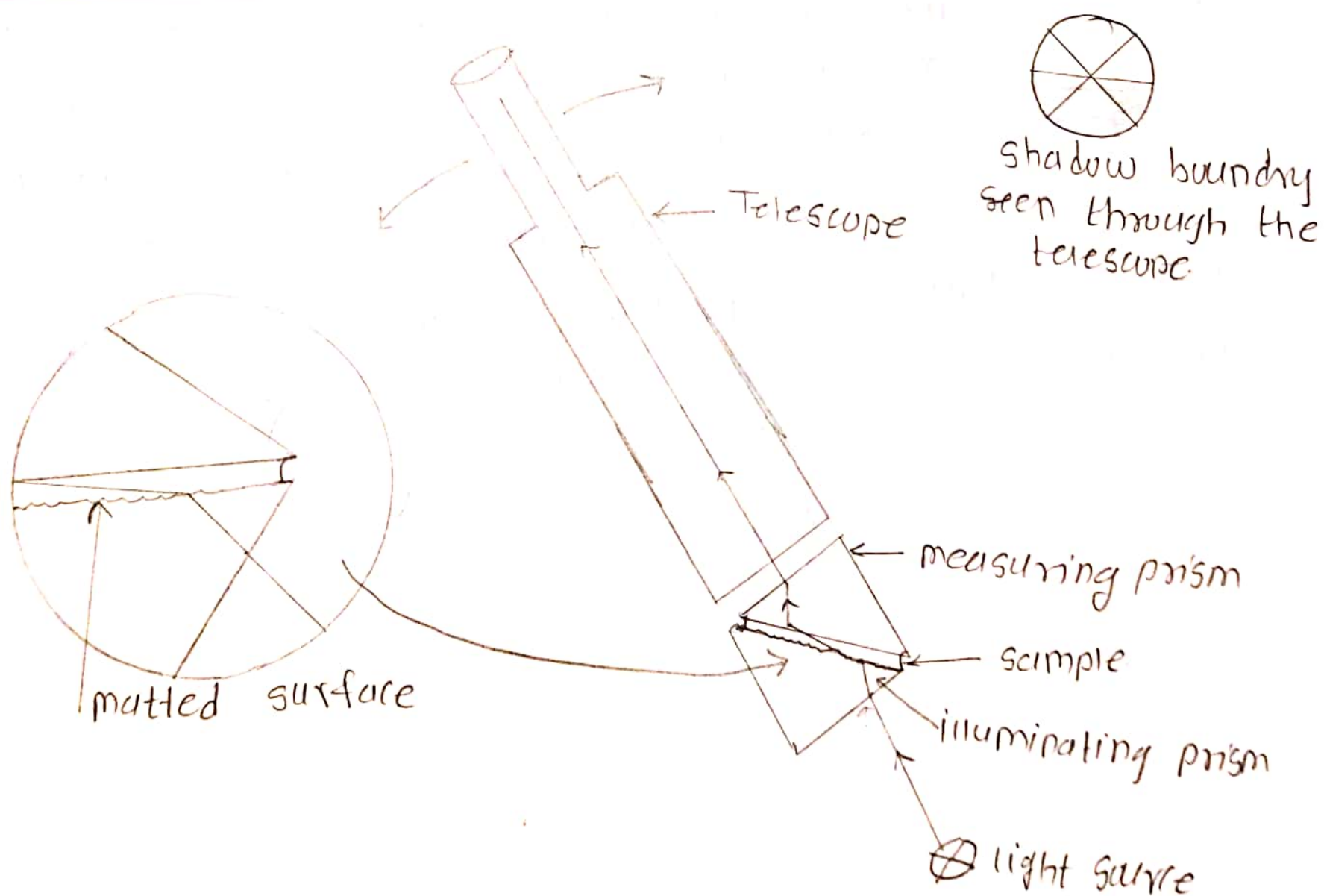
Unit-6

Refractometer.

Principle

Refraction is nothing but bending of light rays when they cross the boundary of media having different densities. Every substance has a physical characteristics that causes refraction of light coming through air. The ability of every substance to refract the light is express in terms of refractive index (R.I.). Since every substance has it's characteristics ~~mean~~ R.I. The concentration of know components in gas or liquid mixture can be determine by measuring the R.I. of the mixture. There exists a linear relationship between R.I. & concentration of two components in the mixture.

Refractometer:



fluid outlet

feed in
ultraviolet

- when the light ray crosses the boundary from denser medium to rarer medium its angle of refraction is greater than angle of incidence.
- This continues until at certain angle of incidence called the critical angle, the refracted ray emerges along the boundary between the two media. If angle of incidence is increased beyond the critical angle, the refraction does not take place, but the ray gets totally internally reflected with maximum intensity.
- The R.F. of the sample solution with respect to glass is measured in terms of the critical angle for this glass-liquid boundary that varies with the composition of the liquid.
- Refractometer measures the R.F. of the sample liquid in terms of its critical angle from which the composition of liquid can be determined when telescope is directed along the critical angle towards the interface, then reading is taken by focussing on sharp division between dark and light portions in the field of view.

These
8000
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Unit-6 ultra violet (UV) absorption Spectroscopy

These spectrometers operate in the range from 2000 Å to 8000 Å which actually includes the visible range also. The basic UV absorption analyser consists of a radiation source, optical fiber filter, a sample cell, a detector & output meter. The absorbance of a substance is given by Beer's law. written as

$$A = abc$$

where- a = molar absorptivity
 b = Path length
 c = Concentration

7) UV Source: The source provides the desired UV wavelength and may be either a line source such as mercury arc or a continuous source such as hydrogen.

8) Optical Filter: It is a spectral dispersing system used for screening out radiations of unwanted wavelengths emitted by the source.

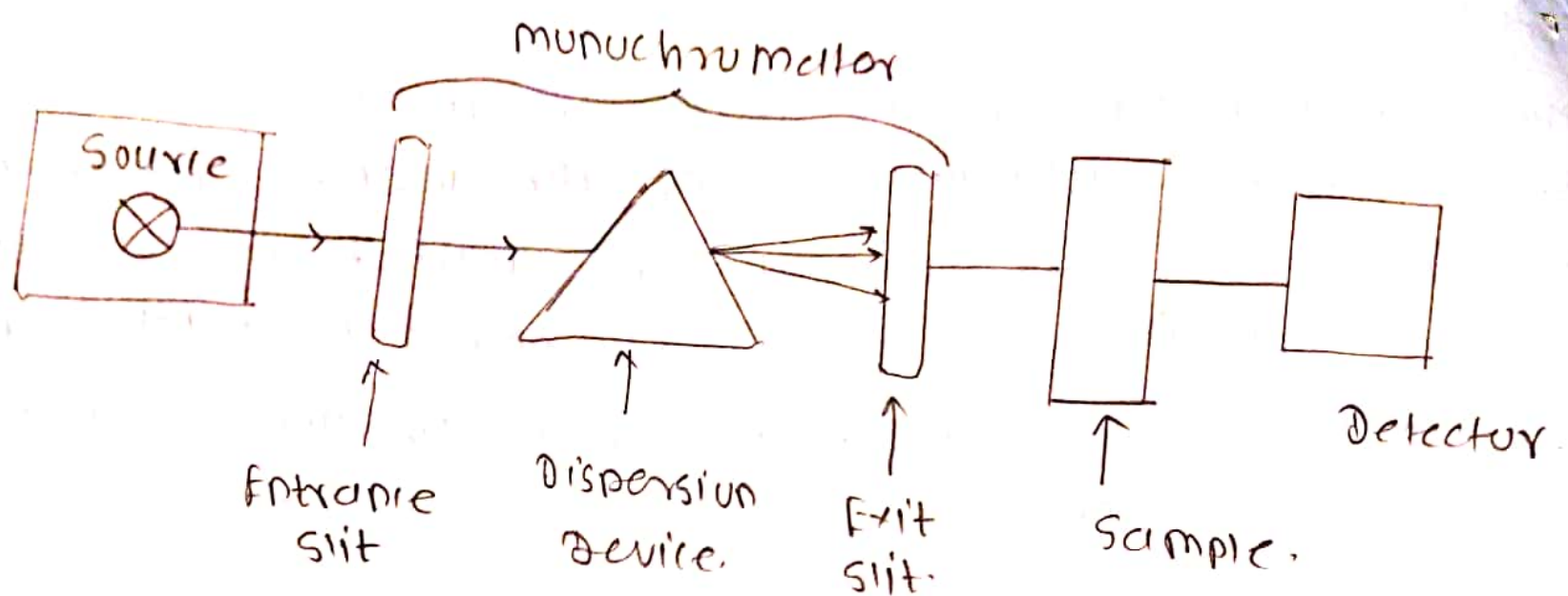
Sample cell: The cell is equip with windows that are transparent at the chosen wavelength.

Detectors: vacuum phototubes photomultipliers and solid state cells are used as UV detectors and their output is read with sensitive meter.

The different type of UV absorption analyzers are:

- ① single beam analyser
- ② split beam analysis.

fluid outlet.
 @ single beam analyzer.



In this method, a single UV beam is passed through the cell, but its output is affected by fluctuation of the light source, dirt or bubbles in the sample cell.

⑥ Split beam analysis:

- In this method radiation from the source are partially absorbed while passing through the sample and the radiation leaving the sample are divided into two beams by a semitransparent mirror.
- Each beam pass through an optical filter to a phototube.
- Each phototube develops a current which is converted to d.c. potential and hence the difference between the potentials of two tubes is proportional to the sample concentration.

Applications

- For analyzing vitamin components.
- For analyzing elemental halogens like Cl_2 , F_2
- ethylene & carbonyl compounds